

Research paper

Economic impacts of material selection on check dam construction in Japan: An input–output analysis promoting timber use

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ABSTRACT

Timber is an excellent eco-friendly alternative to concrete; however, the high costs associated with timber limits its application as a construction material. In this study, the economic impact of timber was evaluated as an added value to offset its high construction costs using check dams as target structures. The 2020 input–output table for Japan was used for the input–output analysis of the economic impact. An extended input–output table was developed to compare the economic impacts of timber, concrete, and steel check dam construction. Although the cost of timber check dam construction was approximately 2.0 and 1.7 times those of concrete and steel check dam construction, respectively, the economic impact ratios, which indicate the relative magnitude of economic impacts, were 2.04, 1.83, and 1.92 for timber, concrete, and steel, respectively. Timber check dam construction had an economic impact that was approximately 10 % higher than those of concrete and steel check dam construction. In terms of eco-efficiency, timber check dams had approximately four and three times the effects of concrete and steel check dams, respectively. Although the construction costs of timber check dams are higher than those of concrete and steel check dams, timber construction can positively impact economies. The replacement of conventional concrete and steel check dams with timber check dams can reduce greenhouse gas emissions and revitalize economies. This study promotes the added value of timber and its broader utilization in civil engineering structures, supporting the development of forestry and timber industries.

1. Introduction

The development of a sustainable society that balances both the environment and economy is attracting significant attention to mitigate the effects of environmental issues, such as climate change caused by global warming and the effect of expanding economic activities. Although housing and infrastructure are essential for economic development, the building and construction sector accounts for approximately 34 % of global greenhouse gas (GHG) emissions in 2023 [1]. Among construction materials, concrete and steel are in high demand worldwide and emit substantial amounts of GHGs throughout their life cycle, including production, transportation, and use. Although efforts are being made to reduce GHG emissions in the cement, concrete, and steel industries [2–7], mitigating emissions from the broader building and construction sector remains a significant challenge [8–13]. The Sixth Report of the Intergovernmental Panel on Climate Change stated that to achieve emission reductions in cities, dwellings, and infrastructure, low-emission building materials can be used in constructing

buildings and recycling building materials to enhance carbon uptake and storage in the urban environment. Examples of low-emission building materials include bio-based building materials, permeable surfaces, and urban green and blue infrastructure [14]. Material Economics estimates that a circular economy approach can reduce GHG emissions by 38 % in the building sector by 2050 by reducing the demand for key materials, such as aluminum, steel, plastics, and cement [15]. Therefore, timber is being promoted as a sustainable alternative material for use in the construction sector [16–21]. The replacement of concrete and steel with timber reduces GHG emissions and mitigates the effects of global warming [22–24]. From a technical perspective, numerous studies have been conducted on the use of timber, and its structural performance and durability have been reported extensively [25,26]. In recent years, countries such as Canada and members of the European Union have actively promoted timber construction as part of their climate-change mitigation strategies. Timber use is expected to contribute to the reduction of GHG emissions and the revitalization of regional economies. These countries have implemented multifaceted

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measures, including policy incentives, the development of technical standards, and the promotion of timber use in public procurement processes [27–30]. In contrast, Japan's adoption of timber in civil engineering structures remains limited and lags behind international trends [31]. Previous research on the economic impact of timber use often relied on static input–output models, which fail to capture dynamic inter-industry effects and regional economic feedback. Moreover, detailed analyses of the economic impacts resulting from timber substitution in construction materials are scarce, hindering a comprehensive understanding of its broader economic significance. Furthermore, concrete remains a relatively inexpensive construction material, and replacing it with timber, which is generally costlier, presents a challenge in terms of increased construction costs. Such economic barriers are among the factors hindering the broader adoption of timber in construction.

Hassan et al. [32] compared the costs of using cross-laminated timber (CLT) and concrete slabs for floor construction in buildings and reported that CLT was more expensive than concrete slabs. Ahmed and Arocho [33] compared the costs of constructing high-rise buildings using concrete and mass timber and found that mass timber was more expensive than concrete because of the higher costs associated with engineered wood production, timber installation, and project staffing. Gu et al. [34] compared the life-cycle costs of mass timber and concrete as construction materials and revealed that although mass timber was more expensive in terms of construction costs, mass timber had lower life-cycle costs. The cost advantage of mass timber can be attributed to its longer lifespan than that of concrete. When comparing costs over lifespans, concrete becomes less expensive than timber. Iwase et al. [35] and Noda et al. [36] demonstrated that substituting concrete with timber as a construction material results in a reduction of GHG emissions; however, this results in an associated increase in construction costs.

The impact of timber substitution on the regional economy has been investigated, highlighting the role of timber as an added value that offsets construction costs. Scouse et al. [37] reported that the use of timber in high-rise buildings can stimulate the regional economy. They highlighted the economic and employment benefits of using locally manufactured mass-timber products, such as CLT, for high-rise construction. Žemaitis et al. [38] compared timber and concrete-based building materials and observed that the use of timber significantly enhances regional employment, wages, and salaries. They observed that the use of timber as a building material contributes to achieving sustainable regional economy.

In Japan, the demand for timber in the architecture sector, which has been the largest consumer of timber, is declining because of the shrinking population [39]. The use of timber in the civil engineering sector is attracting considerable attention [39,40,31]. In this study, we focused on the use of timber in civil engineering structures and evaluated its regional economic impact as a value-added alternative to construction cost. Although the economic impacts of timber use in buildings [41–44,37,38] have been evaluated, few studies have focused on civil engineering structures. In a study on civil engineering structures, Huzita et al. [45] compared the regional economic impacts of CLT and concrete slabs on small-scale bridge decks. However, the scope of their study was limited to small-scale bridges, and reports on general road bridges are scarce. Huzita et al. [46] and Huzita and Kayo [47] compared the regional economic impacts of timber and concrete check dams by redesigning already constructed check dams to obtain the same sediment runoff prevention capacity. However, the check dams were located at different installation sites in different environments and were designed in different years. Moreover, the analysis was limited to Akita Prefecture. Because the evaluation was not conducted at the design stage, challenges were encountered in assessing the added value of timber use in terms of its potential economic impact at the time of the construction method selection. Consequently, comparing their characteristics was challenging. Therefore, this study quantitatively evaluated the added value of timber use by investigating the economic impacts at

the design stage for check dams based on a previous study [48]. By designing timber, concrete, and steel check dams in the same installation environments and with identical functions, we compared the economic impacts under conditions in which functional differences did not exist among check dams. This study evaluated the added value of the economic impact of structures constructed using different materials to select appropriate construction types based on factors other than construction costs. The economic impacts of timber in civil engineering structures were compared with those of other materials.

2. Materials and methods

2.1. Check dams investigated in the study

Based on a previous study [48], check dams were the target structures. A check dam is a civil engineering structure used in forestry to mitigate erosion in streams and to stabilize sediments originating from upstream areas. Consistent with previous studies, this study investigated check dams constructed using timber, concrete, and steel. Table 1 lists the dimensions of the check dams. Sometimes, timber check dams have a smaller height and length than concrete and steel check dams. However, the three types of check dams examined in this study are designed to prevent the same sediment retention capacity, indicating that there are no significant differences in their fundamental functions. These check dams are designed based on design standards [49,50], with their structures adjusted to meet performance requirements according to sediment retention capacity and watershed conditions. Therefore, even if dimensional differences exist due to variations in structural materials, their functional performance is equivalent and comparable.

2.2. Input–output analysis

The methodology employed in this study was based on the analytical framework developed in our previous studies [47,46].

2.2.1. Scope of evaluation

The input–output table used in this study was the input–output table for Japan [51]. The input–output analysis encompassed direct, indirect, secondary, and total effects. Direct effects pertain to the augmentation in production value within Japan resulting from increased final demand. Indirect effects refer to the level of inducement generated in each industry where production increased owing to direct effects. Secondary effects represent the induced value generated through the consumption of services and goods in each industry, attributable to the employment income derived from the direct and indirect effects. The total effect is the aggregate of the direct, indirect, and secondary effects. The economic impacts were assessed in terms of the induced production value and induced gross value added for each effect. The induced production value signifies the output induced in each industry to accommodate the final increase in demand generated by a specific industry. The induced gross value refers to the added value generated from new production owing to the increase in final demand and represents the gross value added, contributing to the national gross domestic product.

In this study, the input coefficients for each industry were assumed to be constant. Furthermore, variations in these coefficients owing to price fluctuations and technological innovation were neglected, and it was assumed that supply could immediately respond to changes in demand.

Table 1
Scales of check dams [48].

Check dam	Length (m)	Height (m)	Crown width (m)	Volume (m ³)
Timber	20.7	3.00	1.80	156.6
Concrete	21.0	4.00	1.50	150.5
Steel	22.0	4.00	1.50	154.7

2.2.2. Extended input–output table

Extended input–output tables were prepared based on previous studies [47,46]. The extended input–output table was based on the existing 2020 input–output table (108 sectors) [51], with the addition of a sector for the construction of check dams, resulting in 109 sectors. The extended input–output tables were developed for the construction of timber, concrete, and steel check dams.

The unit prices of materials (timber, concrete, steel, etc.) used in the construction of the check dams were estimated and determined by the contractor for each designed check dam. Each expense item in the design document of the check dam was classified into the corresponding industry sector using the 2020 input–output Table for Japan Sector Classification [51]. Each expense item in the design document was categorized into direct construction, indirect construction, or general administrative costs. Direct construction costs included material, labor, and direct costs, and the breakdown of the costs was detailed in the design document. Therefore, each expense item was classified into a corresponding sector. However, the indirect construction and general administrative costs were calculated as a percentage of the total construction cost, and their breakdown varied from project to project. Therefore, the details of the breakdown are unknown, which limits sector classification. Interviews with the concerned parties revealed that most of the indirect construction and general administrative costs were included in the gross value-added sector, such as labor costs for the clerical staff of the operators and operating surpluses. Hence, indirect construction costs and general administrative costs were classified under the gross value-added category.

The check dams were designed for 2024, and the input–output table reflected the industrial structure in 2020. Therefore, we multiplied the construction material price index [52] by the price of each cost item and converted the prices in the design document to their 2020 equivalent. Purchaser prices, including commercial and transportation margins, were considered design specifications. Producer prices, excluding commercial and transportation margins, were included in the input–output tables. Therefore, converting the purchaser price into a producer price in the design document is necessary. First, commercial and domestic freight rates were derived from the classification of input tables [51]. The producer price was subsequently calculated by applying the commercial margin and domestic freight rates to the purchaser price in the design document. The eliminated margins were classified into

corresponding commercial and transportation sectors.

Table 2 lists producer prices (in 2020) and intermediate demand–input sectors for each dam construction sector. Fig 1 shows a schematic of the extended input–output table.

2.2.3. Economic impact

Fig 2 illustrates the procedure for evaluating economic impacts. The input coefficient matrix (A) and inverse coefficient matrix (B) were calculated using the following equations [54]:

$$a_{ij} = \frac{x_{ij}}{X_j} \quad (1)$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad (2)$$

$$B = [I - (I - M)A]^{-1} \quad (3)$$

where i denotes the row sector number, j denotes the column sector number, a_{ij} is the input coefficient, x_{ij} is the input from sector i to sector j , and X_j is the in-county output of sector j . I represents the identity matrix, and M is the diagonal matrix of the import coefficients. The import coefficient was calculated by dividing the amount of imports in each sector by the total domestic demand for that sector. However, the import coefficients for the forestry sector and lumber and wood products sector in the timber check-dam construction sector were set to zero because Japanese cedar (*Cryptomeria japonica* D. Don) timber used for the timber check dams was obtained locally, ranging from the collection of the raw materials to their processing into timber. Value-added coefficients were calculated by dividing the gross value added in each industrial sector by the domestic production value. Income coefficients were obtained by dividing the employment income of each industrial sector by the domestic production value. The propensity to consume (0.528698) was obtained by dividing the consumption expenditure by the income in the latest 2023 Annual Report of the Family Income and Expenditure Survey [55]. Personal consumption expenditure coefficients were obtained by dividing the amount of personal consumption expenditure in each sector by the total personal consumption expenditure.

3. Results and discussion

3.1. Economic impacts

Table 3 lists the economic impact of the construction of each type of check dam. The construction cost (final demand increase) obtained from the design documents of each type of check dam revealed that the timber check dams were the most expensive, with costs approximately 2.0 and 1.7 times higher than those of concrete and steel check dams, respectively. The concrete check dam was the least expensive. In architecture and civil engineering, timber is more expensive than concrete because of material, labor, and overhead costs [32,45,37]. The use of timber can result in increased costs during the design modification phase [33]. In this study, the construction of the timber check dams resulted in lower labor costs than those of concrete and steel check dams. Nevertheless, the material and overhead expenses for timber check dams were higher than those for concrete and steel check dams, aligning with previous research findings [32,45,37]. Overhead expenses increase in proportion to construction costs; therefore, when material costs increase, overhead expenses also increase. Interviews with operators and related parties revealed that the timber check dams have high costs because the timber used for their construction did not conform to the sizes commonly distributed in the Japanese market; therefore, timber had to be custom-made, which increased its cost. If timber conforming to commonly available market sizes in Japan can be utilized, the cost of

Table 2

Assignment of expenses to the industrial sector for each check dam construction sector (in Japanese yen).

Industrial sector	Check dam construction sector		
	Timber	Concrete	Steel
Crop cultivation	112,967	45,240	24,578
Miscellaneous mining industry	231,476	0	774,483
Textile products	47,433	431	15,381
Lumber and wood products	6223,467	623,709	2911
Petroleum refinery products	72,499	42,875	88,428
Plastic products	308,627	311,282	308,627
Cement and cement products	0	2160,527	0
Steel products	32,712	0	4064,944
Nonferrous metal products	0	35,228	35,228
Miscellaneous metal products	2639,668	0	0
Commerce	1668,589	871,846	433,176
Railway transport	1335	1021	671
Road transport (except self-transport)	590,294	201,246	188,196
Water transport	37,269	26,229	61,067
Air transport	84	17	8
Freight forwarding	32,960	10,597	11,869
Storage facility service	71,495	11,092	52,781
Goods rental and leasing services	770,769	137,093	281,895
Compensation of employees	1249,287	2291,561	1292,341
Gross value added	11,997,031	6021,136	7872,878
Total	26,087,963	12,791,138	15,509,462

Demand sector (buyer) Supply sector (seller)		Intermediate demand							Final demand	(Less) Imports	Prefectural production
		Crop cultivation	Livestock	...	Public construction	Check dam construction	...	Total			
Intermediate demand	Crop cultivation	Column				↓					
	Livestock										
	...										
	Public construction										
	Check dam construction	→	Row								
	...										
	Total										
Gross value added											
Prefectural production											

Fig. 1. Extended input–output table (based on the [53], developed by the author).

timber dam construction may be reduced.

The economic impact (total effect) was the greatest for timber check dam construction because of its highest increase in the final demand. To evaluate the relative economic impacts, we compared them using the economic impact ratio, which is the total effect divided by the increase in the final demand. Among the three types of check dams analyzed in this study, the economic impact ratio was the highest for the timber check dam (2.04), followed by the steel check dam (1.92) and concrete check dam (1.83). The construction cost and economic impact of the timber check dam were higher than those of the concrete and steel check dams. Although timber check dams are more expensive to construct than concrete and steel check dams, their economic impact is approximately 10 % higher than those of the concrete and steel check dams. Although the total number of check dams constructed annually in Japan is unknown, it is estimated that a significant number of check dams have been constructed nationwide, given that the budget allocated for mountain disaster prevention projects by the Forestry Agency in fiscal year 2024 reached 62.5 billion yen [56]. Previous research has shown that timber check dams can be used as alternatives to concrete check dams <5 m high [31]. The same study estimated the annual construction of concrete check dams with heights smaller than 5 m to be 980 units [40], assuming a 50 % or 100 % replacement rate. If 490 units (50 %

could be replaced with timber check dams each year, the economic impacts of timber check dams would be approximately 10 % higher than those of concrete check dams. Therefore, in regions with active forestry industries, the expansion of timber demand would likely contribute to the promotion of forestry.

The results of this study reveal that promoting timber use in civil engineering can revitalize economies. In Japan, subsidy programs are available for the use of domestic timber [57]. The recognized added value of timber check dams in terms of economic impact may justify their relatively higher construction costs, particularly when combined with subsidized projects.

Timber check dams generate higher economic impacts than concrete and steel check dams. However, from a technical perspective, careful consideration is required regarding the durability and long-term maintenance of timber. Reports on the durability and maintenance of timber check dams [58–61] highlight potential challenges, such as the need to consult local engineers for periodic maintenance and to establish necessary maintenance systems. In contrast, some timber check dams have maintained their flood control functions for >60 years [62], suggesting that long-term functional maintenance is possible with appropriate design, construction, and regular maintenance. Therefore, when introducing timber check dams, their economic effects and other factors,

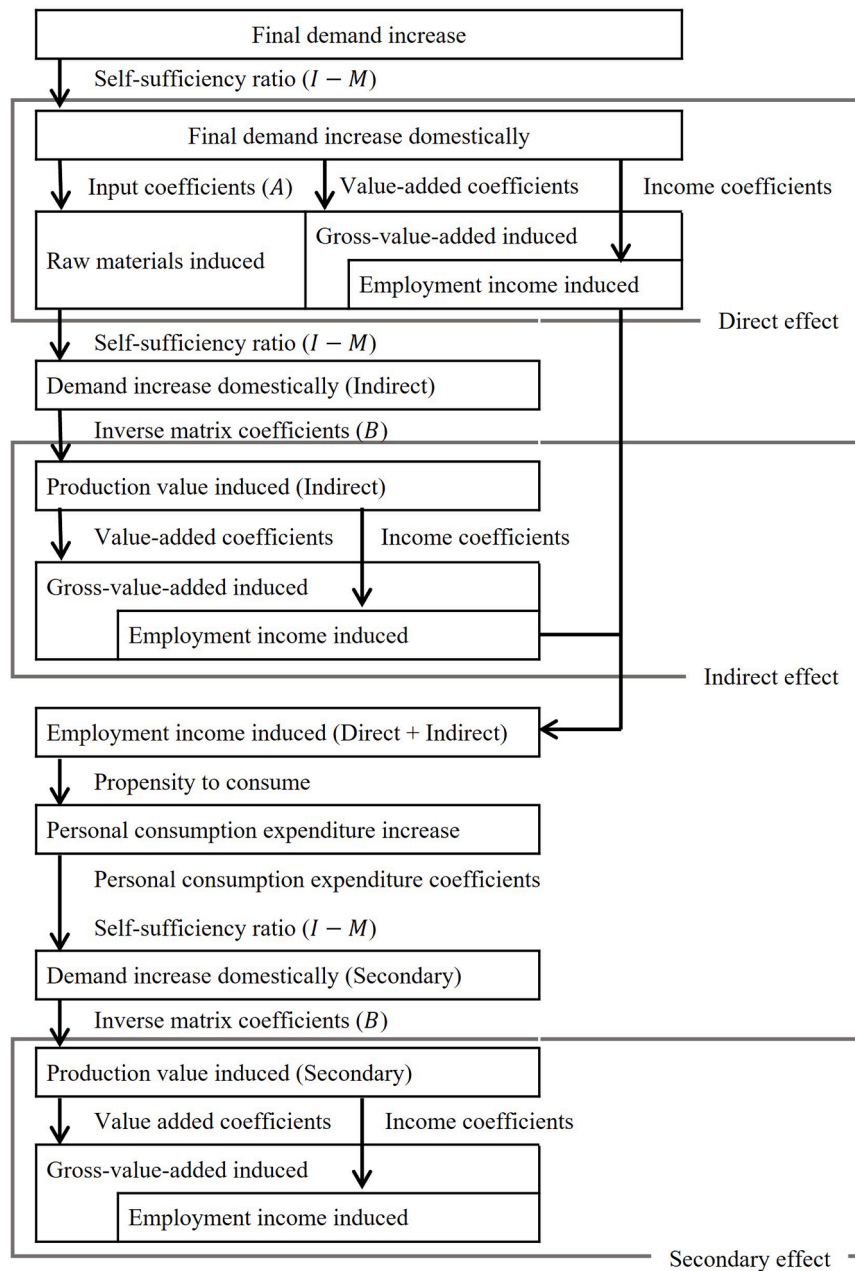


Fig. 2. Procedures for assessing economic impacts (based on Huzita and Kayo [47], developed by the author).

such as lifespan, maintenance cycles, and the technical capabilities of the region, must be evaluated comprehensively.

Fig 3 (Table S1) lists the top five sectors with indirect effects for each check dam. Timber check dam construction significantly impacts the lumber and wood products sector and its upstream forestry sector. Concrete check dam construction significantly impacts the cement and cement product sectors, whereas steel check dam construction significantly impacts the steel products and pig iron and crude steel sectors.

The construction of each type of check dam significantly impacts the industry sectors that manufacture essential materials. The lumber and wood products and forestry sectors played a dominant role in generating economic impacts, accounting for approximately 39 % of the indirect effects associated with timber check dam construction. Moreover, the use of timber as the main body of check dams significantly impacted the timber industry, encompassing the forestry industry. Because some reports reveal that the plantation forestry industry contributes to employment and infrastructure development in the region [63], the

impact on the forestry industry may contribute to the promotion of the regional economy. Concrete check dam construction significantly impacted the lumber and wood products sector, which ranked third to the cement and cement products and commerce sectors. However, the impacts on the upstream industry of forestry were insignificant. In Japan, the use of in-place timber formwork is being promoted for the construction of concrete check dams, as opposed to conventional removable forms, to enhance timber utilization and improve on-site workability [64,65]. To date, the economic impact of using timber forms left in place is yet to be examined. The question is whether the economic impact of using timber forms left in place contributes to the promotion of forestry, an upstream industry. However, this study demonstrated the potential of timber use in concrete structures to contribute to the economy using timber forms that remained in place.

Table 3

Economic impact (measured in thousand Japanese yen or in people) of the construction of check dams.

Check dam		Timber	Concrete	Steel
Final Demand Increase (F)		26,088	12,791	15,509
Direct effect	Production value induced	26,088	12,791	15,509
	Gross-value-added induced	13,246	8313	9165
Indirect effect	Production value induced	22,278	7493	12,207
	Gross-value-added induced	11,213	3930	4135
Secondary effect	Production value induced	4994	3117	2123
	Gross-value-added induced	2932	1830	1246
Total effect	Production value induced (T)	53,360	23,402	29,840
	Gross-value-added induced	27,391	14,072	14,547
Economic impact ratio	T/F	2.04	1.83	1.92

3.2. Sensitivity analysis

In this study, the timber used for constructing the timber check dams were sourced locally. Therefore, the import coefficients for the forestry sector and the wood and wood products sector was set to zero (indicating a self-sufficiency rate of 100 %). Currently, timber check dams are made from timber processed from special orders; hence, the timber is sourced locally. However, in the future, it may become possible to use timber that is readily available on the market. Considering that Japan's timber self-sufficiency rate is 40.7 % [56], it may be impossible to satisfy all needs using locally produced timber. Therefore, we evaluated whether the economic impact of using timber remains advantageous compared to other materials, even if the self-sufficiency rate of timber changes.

The timber check-dam construction project targeted in this study (self-sufficiency rate: forestry = 100 %; lumber and wood products = 100 %) was designated as Scenario 1. In Scenario 2, we applied the self-sufficiency rates for the forestry and lumber and wood products sectors used in the 2020 input–output table (self-sufficiency rate: forestry = 88 %; lumber and wood products = 71 %) [51]. In Scenario 3, Japan's 2022 timber self-sufficiency rate (40.7 %) was applied to the lumber and wood products sector (self-sufficiency rate: forestry = 88 %; lumber and wood products = 40.7 %). The results for each scenario are presented in Fig. 4.

In Scenario 2, the economic impact of the timber check dam was similar to that of the concrete check dam and lower than that of the steel check dam. In Scenario 3, the economic impact was lower than those of the concrete and steel check dams. Currently, the use of timber in check dams has an economic impact advantage over other materials. However, the supply of domestic timber will be insufficient if commonly available timber becomes usable. This might necessitate the use of imported timber, and the economic impact advantage will be lost. A reason for the stagnation in Japan's timber self-sufficiency rate is the decline of the Japanese forestry industry. This is because forestry occupations are physically demanding, low-paying, and dangerous, which make it difficult to employ and retain new workers. Even if the construction cost of check dams decreases in the future because of the use of readily available timber, the economic impact may be low if the self-sufficiency rate remains low. Policy initiatives, such as improving the wages and working conditions of workers in the forestry sector, are also necessary to maintain the economic impact of timber.

3.3. Eco-efficiency

Eco-efficiency is defined as the economic-added value by a company in relation to its environmental impact [66]. In a previous study [48], the GHG emissions associated with the construction of the check dams examined in this study were assessed through an input–output analysis. Consequently, the eco-efficiency of constructing each type of check dam was determined by dividing the gross value added generated via the construction of the check dam by the corresponding GHG emissions. Table 4 presents the gross value added, GHG emissions, and eco-efficiency related to the construction of each check dam. The eco-efficiency for timber check dam construction was approximately 3.95 and 2.89 times higher than those of the concrete and steel check dams, respectively. These findings suggest that the construction of a timber check dam can optimize economic value addition while simultaneously reducing GHG emissions compared to the construction of other types of check dams.

Kim et al. [67] revealed that a trade-off exists between GHG emission reductions and economic growth, as measured by the gross domestic product. Although a case study, the results of this study quantitatively demonstrated the potential for both GHG emission reduction and increase in the gross value added by replacing conventional concrete and steel check dams with timber check dams.

In Japan, check dams are predominantly constructed using concrete. A previous study [31] estimated that for check dams with a height of 5 m or less, it is feasible to replace concrete structures with timber alternatives. On the basis of these estimates, the construction material of approximately 490–980 concrete check dams can potentially be replaced with timber. If such replacements are implemented on the estimated scale, GHG emission reduction and positive economic impacts on the timber industry, including forestry, can be expected. However, the potential economic impact on the concrete industry resulting from increased timber adoption should be considered.

Although production is curtailed by material substitution to minimize GHG emissions, promoting the use of timber as a less environmentally hazardous material in the construction sector can simultaneously reduce environmental impact and revitalize economy.

3.4. Comparison with previous studies

In previous studies [47,46], economic impacts were evaluated using data collected from project sponsors, including actual expenditures on construction materials and labor. However, post-construction assessments require data that have already been used, making it impossible to select construction types while considering economic impacts in advance. In contrast, this study conducted the evaluation during the pre-construction design phase. A comparison with previous studies was conducted to analyze the differences between evaluations based on actual financial data (post-construction data) and those based on design-stage data.

Although previous studies [47,46] assessed the economic impacts of similar timber check dams in Akita Prefecture, the dams differed in terms of design year and dam size. Table 5 presents the findings of the comparative analysis of economic impacts between these previous studies and this study. Huzita et al. [46] assessed the economic impact of a timber check dam constructed in 2013 using the 2011 input–output table. Huzita and Kayo [47] used the same 2015 input–output table used in this study to evaluate the economic impacts of a timber check dam constructed in 2013. The 2013 and 2024 models were approximately two times as expensive to construct than their concrete counterparts. The tendency of the timber check dam to exhibit higher economic impacts than the concrete check dam is consistent with the results of previous studies. Despite the difference in the target regions—Akita Prefecture in particular and Japan in general—no significant variation was observed in the trend of economic impacts. This indicates that evaluations based on data from the pre-construction design stage can

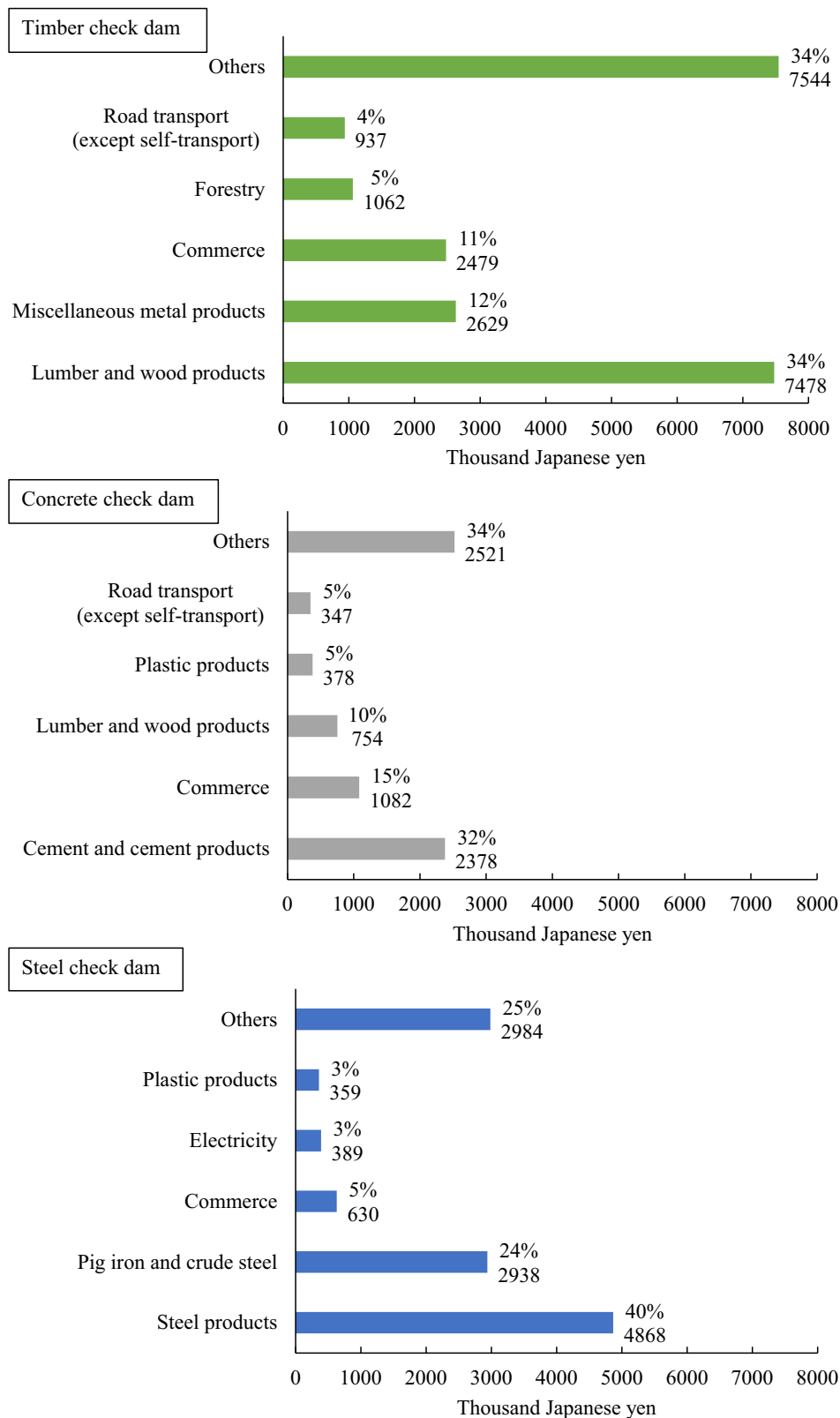


Fig. 3. Indirect effect of economic impact of the construction of check dams: top five sectors.

yield results comparable to those of previous studies. These findings suggest that incorporating economic impact assessments at the design stage can support more informed decision-making in public works projects, thereby promoting the adoption of timber structures.

3.5. Limitations of the study

This study focused on check dams designed in Akita Prefecture, Japan. In general, the prices of construction materials in Japan show limited regional variations. In addition, material prices were

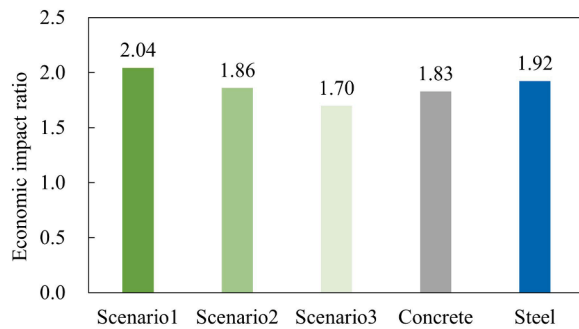


Fig. 4. Sensitivity analysis results. scenario 1 (Timber check dam, self-sufficiency rate: forestry = 100 %; lumber and wood products = 100 %); scenario 2 (self-sufficiency rate: forestry = 88 %; lumber and wood products = 71 %); scenario 3 (self-sufficiency rate: forestry = 88 %; lumber and wood products = 40.7 %), concrete: concrete check dam; Steel: Steel check dam.

Table 4
Eco-efficiency of check dams.

Check dam	Timber	Concrete	Steel
Gross-value-added induced (Total effect) (thousand yen)	27,391	14,072	14,547
GHG emission (kg-CO ₂ eq) [48]	24,128	49,022	35,801
Eco-efficiency (thousand yen/ kg-CO ₂ eq)	1.1352	0.28705	0.39306

standardized in this study by converting them to national averages. Therefore, the use of the national input–output table of Japan is considered appropriate, even for evaluating check dams constructed in different regions. However, previous studies in Japan have shown that regional variations in the prices of ready-mixed concrete can be significant [68], and that labor costs may differ considerably by region [69].

In this study, the economic impacts of check dams designed in 2024 were evaluated using the input–output table that reflected the industrial structure of 2020. Because the input–output table is typically updated every five years, the latest available version as of 2025 is based on the 2020 industrial structure, which may differ from the current economic conditions. Previous studies have attempted to estimate economic impacts by forecasting future industrial structures. However, such predictions are inherently limited owing to unforeseeable factors, such as technological innovation and the emergence of new industrial sectors [45]. Therefore, addressing the structural gap between the input–output table and the year of project design is crucial.

The economic impacts evaluated in this study were limited to the construction of check dams. In Japan, concrete is the main material used for check dam construction, and minimal maintenance and removal are performed. However, in the case of timber check dams, regular maintenance is necessary because of concerns regarding durability and long-term performance. Therefore, comprehensive evaluations that include the cascade use of timber should be conducted. It is necessary to focus on the life cycle of check dams and evaluate their impact on downstream industries, taking recycling into consideration.

Table 5
Comparison with previous studies on economic impacts of the construction of check dams.

Check dam	Design year	Data type	IOT ^a year	Cost (thousand yen)	Economic impact ratio	Reference
Timber	2013	post-construction	2011	30,000	1.94	[46]
Timber	2013	post-construction	2015	31,001	1.86	[47]
Timber	2024	design-stage	2020	26,088	2.04	This study
Concrete	2018	post-construction	2011	15,556	1.72	[46]
Concrete	2018	post-construction	2015	14,595	1.59	[47]
Concrete	2022	design-stage	2020	12,791	1.83	This study

^a IOT denotes “Input–Output Table.”.

4. Policy implications

The use of timber in check dams is limited by challenges, such as durability [70,60], and timber decay. In addition, the high initial cost of using timber and economic uncertainty are barriers to its introduction [71]. Therefore, a roadmap is proposed to promote the use of timber check dams (Table 6). This roadmap aims to implement the use of timber in society through gradual institutional reforms and technological development.

For the short-term goal, provisional design guidelines should be established, and a model timber check dam should be designed to address the non-standardization of design codes among prefectures and the lack of unified standards. For the medium-term goal, the design standards established in the short-term goal should be formalized. Additionally, a subsidy system should be established for timber utilization to address economic uncertainty and promote economic incentives for timber use. Furthermore, the supply system should be strengthened to accommodate increased timber demand. For the long-term goal, the policy should be expanded nationwide, and efforts need to be made to integrate it with national and regional forest policies and climate change countermeasures, aiming to expand the use of timber beyond check dams. The proposed roadmap is expected to enhance the sustainability of forest conservation projects and contribute to the promotion of forestry, including the timber industry.

5. Conclusions

In this study, an extended input–output table was created using design documents to compare the economic impacts of timber check dam construction with those of concrete and steel check dam construction at the design stage. The conclusions of this study are as follows.

- (1) The economic impact ratios of the check dams were 2.04, 1.83, and 1.92 for timber, concrete, and steel, respectively. These values indicate that timber check dam construction yielded the highest economic impact.
- (2) Although the timber check dam is more expensive to construct than concrete and steel check dams, their economic impact is approximately 10 % higher than those of the concrete and steel check dams.
- (3) The construction of timber check dams significantly impacts timber industries, including forestry, because of the use of timber as the main construction material. Additionally, using wooden

Table 6
Proposal for a policy roadmap.

Period	Main Policies
Short-term (approximately three years)	Establish provisional design guidelines; Design a model timber check dam
Medium-term (approximately six years)	Formalization of design standards; Development of subsidy systems; strengthening of the regional timber-supply system
Long-term (more than seven years)	Nationwide expansion; Integration with forest policies and climate-change countermeasures

forms left in place may contribute to the revitalization of timber industries, even for concrete check dams.

- (4) The replacement of concrete and steel check dams with timber check dams can simultaneously reduce GHG emissions and revitalize the economy.

The findings of this study support the promotion of timber use in civil engineering structures. In addition to using timber as the primary construction material of the main structure, we verified the economic impacts of using wooden forms left in place. This study can be applied to evaluate the economic impacts of using timber in both the main structures and other components of civil engineering projects. By assessing the economic impacts during the design phase, the selection of timber is justified despite its higher cost, as it provides additional value beyond the economic cost. Future studies should consider regional differences in material and labor costs. In addition, the life cycle assessment of check dam construction should be conducted and evaluated together with the economic impacts. This approach will enhance the repertoire of added value and expand the options for timber selection.

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CRediT authorship contribution statement

Tomohumi Huzita: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ryu Noda:** Writing – review & editing, Validation, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Tomohumi Huzita reports article publishing charges, equipment, drugs, or supplies, and travel were provided by Yanmar Environmental Sustainability Support Association. Tomohumi Huzita reports article publishing charges, equipment, drugs, or supplies, and travel were provided by Japan Science and Technology Agency. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

Data will be made available on request.

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